Student Guide — middle / module-12-movement-of-sun

IKS Curriculum

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Student Workbook

Student Workbook — Movement of the Sun (Middle)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

Vocabulary tracker

Fill in as we go:

- *Sūrya* — ___________________________
- *Uttarāyaṇa* — ___________________________
- *Dakṣiṇāyana* — ___________________________
- *saṁkrānti* — ___________________________
- *rāśi* — ___________________________
- *sāyana* — ___________________________
- *nirayana* — ___________________________

Day-by-day notes

Day 1 — The Sun's Yearly Path

What I learned:

One question I have:

Day 2 — Axial Tilt and the Seasons

What I learned:

One question I have:

Day 3 — Uttarāyaṇa and Dakshiṇāyana

What I learned:

One question I have:

Day 4 — Equinoxes — Viṣuva

What I learned:

One question I have:

Day 5 — Twelve Rāśis

What I learned:

One question I have:

Day 6 — Saṁkrānti Moments

What I learned:

One question I have:

Day 7 — Sāyana vs Nirayana

What I learned:

One question I have:

Day 8 — Precession of the Equinoxes

What I learned:

One question I have:

Day 9 — Cross-Cultural Solar Calendars

What I learned:

One question I have:

Day 10 — Final Share and Assessment

What I learned:

One question I have:

Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

Day 1 — What is the Sun Doing?

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will describe the Sun’s daily and annual apparent motion and articulate the question: *does the Sun rise from exactly the same spot every day?*

Materials

- Whiteboard
- A 30 cm wooden dowel or pencil + a piece of cardboard / paper to plant it on (the gnomon)
- A compass (or a phone with a compass app)
- A printout of a year-calendar (Jan to Dec, all dates visible) — one per pair
- Optional: a photo of the room taken at the same time on different days (a teacher prep item)

Vocabulary introduced today

- *Sūrya* (IAST: *sūrya*; lit. “the Sun”) — the luminary at the centre of our solar system; the object we will track for the next 10 days.
- *krānti-vṛtta* (IAST: *krānti-vṛtta*; lit. “deflection-circle”) — the **ecliptic**, the apparent path the Sun traces across the sky over a year.

Lesson flow

Warm-up (5 min)

- Prompt on board: *“Where was the Sun this morning when you left home?”*
- Take 3–4 responses. Most will say “east” or “in front of me / behind me.” Some will say “I didn’t notice.” That’s the honest answer — and the launching point.
- Teacher: *“Today we’re going to start noticing. For two weeks we’ll track the Sun. The Indian astronomical tradition has been doing exactly this for 2000+ years.”*

Core (20 min)

1. **Two motions, one Sun.** Sketch on the board:
    - **Daily motion:** rises east-ish, climbs, sets west-ish. Takes ~24 hours.
    - **Annual motion:** the rising spot shifts north (summer) and south (winter) along the horizon, completing a full cycle in ~365.25 days.
2. **The Sun’s apparent path on the sky.** Show this:

```
    NORTH
      |
W---+---E   ← in March/September the Sun rises due east
      |        and sets due west
    SOUTH

W---+---E   ← in June the rising point shifts NORTH of east
              (high arc in the sky, long days)

W---+---E   ← in December it shifts SOUTH of east
              (low arc, short days)
```

- The path the Sun traces against the *stars* over a year is called the **ecliptic** (*krānti-vṛtta*).
- From Earth, we don't see the stars in daytime — but Indian astronomers worked out the ecliptic by noting which stars came up just before sunrise and just after sunset.

3. **Two questions for the module.** Write on the board:
    - **Q1:** Why does the rising spot shift? *(Answer = axial tilt — Day 2.)*
    - **Q2:** What did Indian astronomers call the Sun's slow north-south swing? *(Answer = uttarāyaṇa and dakṣiṇāyana — Day 3.)*

Activity (15 min) — Shadow-stick observation

If the weather permits and the schoolyard has a sunny spot:

- Plant the gnomon (the dowel) upright in a piece of cardboard. Mark north on the cardboard using the compass.
- At a known time (say, the start of the lesson), trace the shadow of the gnomon onto the cardboard. Write the time and the date next to the line.
- Return to the same gnomon in 10 minutes. The shadow has moved. Trace the new line.
- Discuss: *"The Sun seemed to stand still while you were in class — but the shadow moved. What does that tell you about how the Sun moved?"*

If weather doesn't permit: do the activity indoors with a strong torch (acting as the Sun) and a pencil-on-paper gnomon. Move the torch slowly along an arc to mimic the daily motion. Each "minute" of torch travel = ~4 minutes of real Sun travel.

Safety reminder: the students look at the **shadow**, not the Sun. Anyone caught glancing up at the Sun pays a forfeit (a fun one — like doing the daily prompt next class).

Wrap-up (5 min)

- Exit prompt: *"In one sentence — how can you tell tomorrow morning whether the Sun has 'moved north' or 'moved south' since today?"*
- Take 2 responses. Acceptable answer: *"Compare the sunrise spot to a known landmark on the horizon — has the rising point moved closer to north or to south?"*

- Preview Day 2: *“Tomorrow — WHY does the Sun’s spot shift? Bring a globe if you have a small one at home.”*

Student-facing content

The Sun has two motions you can see:

- **Daily** — rises east-ish, sets west-ish. Takes 24 hours.
- **Annual** — the rising spot drifts north and south along the horizon over a year.

The Sun’s yearly path across the stars is called the **ecliptic** (*krānti-vṛtta*). Indian astronomers, like astronomers everywhere, traced this path by watching which stars were just visible at dawn and dusk.

Today, you have noticed the daily motion (via the gnomon). Over the next two weeks, you’ll learn the names for the annual motion: *uttarāyaṇa* (northward course) and *dakṣiṇāyana* (southward course).

Homework

- For three mornings this week, when you leave home, glance at the eastern horizon. Note where exactly the Sun rises relative to a landmark (a tree, a building, a flagpole). Sketch the spot on a small map — three sketches.
- Bring the calendar printout to Day 2.

Differentiation

- **One tier up:** Find out (online or in the school library) the **time** of sunrise and sunset for today AND for one day exactly six months from now. By how many minutes does the day length differ?
- **One tier down:** Just do the morning-glance — don’t worry about sketching maps. Three mental notes are enough.

Teacher notes

- Some students will know the Sun-rises-in-the-east rule and stop there. Push: *“Always east? Or east-ish?”* The “ish” is the entire module.
- Don’t introduce axial tilt yet. That’s Day 2. Let students sit with the *observation* of the shifting spot before getting the *explanation*.
- The shadow activity is more valuable than the explanation. If you only have 30 minutes, run the activity and trim the Core.

Day 2 — Why Earth has Seasons

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will explain that Earth’s ~23.5° axial tilt — not its distance from the Sun — is what causes the seasons, and demonstrate this with a torch-and-globe model.

Materials

- A small globe (or a ball with N and S marked) — one per group ideally, one as demo minimum
- A strong torch / table lamp (the "Sun")
- A protractor (to verify the tilt angle)
- A blackout / curtained room if possible, OR a hallway
- Student notebooks

Vocabulary introduced today

- **Axial tilt** (obliquity of the ecliptic) — Earth's rotation axis is tilted ~23.5° from a line perpendicular to its orbital plane. This tilt does not change over a year — Earth keeps its tilt pointed in the same direction in space as it goes around the Sun.

(No new Sanskrit term today — yesterday's *krānti-vṛtta* covers it. The axial tilt is what *causes* the ecliptic to be a *circle* on the sky rather than just running along the equator.)

Lesson flow

Warm-up (5 min)

- Prompt: *"Why is it hot in May–June in India and cool in December–January?"*
- Likely wrong answer (very common): *"Because Earth is closer to the Sun in summer and farther in winter."*
- Surprise: Earth is actually slightly *closer* to the Sun in **January** (perihelion ~3 January, 147.1 million km) and slightly *farther* in **July** (aphelion ~4 July, 152.1 million km). India is hot in May–June anyway. So distance is not the answer.
- Real question: *"What else could cause seasons?"*

Core (20 min)

1. **The torch-and-globe demo.** Set up:

    - Place the torch on a desk pointing horizontally.
    - Hold the globe at arm's length, with its axis tilted ~23.5° (use the protractor to check). The N pole should lean either toward or away from the torch.

    Move the globe through four positions, each ~90° around the torch, keeping the axis tilt **fixed in space** (the N pole always points the same direction):

Position	What it represents	Northern Hemisphere	Southern Hemisphere
N pole tilted TOWARD torch	June solstice	Summer (long days)	Winter (short days)
N pole tilted SIDEWAY	September equinox	Day = night	Day = night

Position	What it represents	Northern Hemisphere	Southern Hemisphere
S (1)			
N pole tilted AWAY from torch	December solstice	Winter (short days)	Summer (long days)
N pole tilted SIDEWAYS (2)	March equinox	Day = night	Day = night

2. **What students should see:**
    - When N is tilted toward the torch: the upper half of the globe is brightly lit, and the lit region extends past the N pole. India is well-lit and the Sun is high in the sky. → Summer.
    - When N is tilted away: the upper half is poorly lit, the lit region falls short of the N pole. India is lit at a shallow angle. → Winter.
3. **Two things the tilt does:**
    - **Day length** — when tilted toward, the daytime arc through India is *longer* than the night-time arc. → Longer days in summer.
    - **Angle of sunlight** — sunlight hits a tilted surface at a steeper angle in summer (more concentrated, more heat per square metre) and a shallow angle in winter (more spread out, less heat).
4. **Why does the tilt stay fixed?** Earth's rotation acts like a spinning top — it resists changes in its tilt direction. Over very long times (~25,800 years) the tilt direction wobbles in a slow circle (this is **precession** — Day 8 senior band; we hint at it here only as a footnote).

Activity (15 min) — Build your own tilt model

In pairs, students build a paper "season-checker":

- Take an A4 sheet. Mark a small circle in the centre — that's the Sun.
- Draw Earth's orbit as a big circle around it.
- At four positions on the orbit (top, right, bottom, left), draw a small Earth with its axis tilted to the **right** of vertical (same direction in all four).
- Mark which position is summer in the Northern Hemisphere, which is winter, which are the two equinoxes.
- Label each: June / September / December / March.
- Bonus: mark the Indian subcontinent on each Earth and shade where it's day.

Wrap-up (5 min)

- **Exit ticket on a slip of paper:** "Which is hotter in India in June: the *day length* or the *angle of sunlight* on your skin? Or both? In one sentence."

- Acceptable answer: BOTH — longer days mean more total hours of sunlight, AND the steeper angle means each minute of sunlight delivers more heat per square metre.

Homework

- On the calendar printout, mark the four dates: **~21 March (March equinox), ~21 June (June solstice), ~23 September (September equinox), ~21 December (December solstice).**
- Bring the marked calendar to Day 3.

Student-facing content

> **Earth is tilted.** Its rotation axis is at ~23.5° from straight-up (relative to its orbit around the Sun). And the tilt direction *doesn't change* as Earth orbits.
>
> This single fact causes the seasons:

- When the N pole leans TOWARD the Sun (~June 21) → summer in India.
- When it leans AWAY (~December 21) → winter in India.
- When it leans SIDEWAYS (~March 21 and ~September 23) → day = night everywhere on Earth.

> Surprise fact: Earth is actually *closer* to the Sun in January than in July. Distance is not what causes seasons. Tilt is.

Differentiation

- **One tier up:** What would the seasons be like if Earth's tilt were 0°? (No seasons; equal day/night everywhere all year; equatorial regions hot, polar regions cold permanently.) What if it were 45°? (Extreme seasons — the Arctic Circle would extend down to Delhi's latitude in summer.)
- **One tier down:** Focus on the demo. Just remember: "tilted toward = summer, tilted away = winter, sideways = equinox."

Teacher notes

- The "Earth closer to Sun = summer" misconception is *very* persistent. Highlight that India's June is summer despite Earth being farthest from the Sun in July. The misconception breaks instantly when this is named.
- Some students will ask: *"Then why is India summer in June and not in winter like Australia?"* — Yes, India is in the Northern Hemisphere; Australia is in the Southern. Same Earth tilt, opposite seasons. (Australia is wearing shorts in December, jumpers in June.)
- The demo will fail visually if the room is too bright. If you can't darken the room, use a smaller globe and bring the torch closer (3–4 cm from the globe surface).

Citation(s) used in this lesson

- Modern axial-tilt and orbital values from standard NCERT-level astronomy. No primary-source citation needed at this band.

Day 3 — Uttarāyaṇa and Dakṣiṇāyana

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will name the two solstices, identify the half-years they bookend, and use the Sanskrit terms *uttarāyaṇa* and *dakṣiṇāyana* correctly.

Materials

- The calendar from Day 2 (with the four turning-points marked)
- The whiteboard
- A printable two-circle handout (one circle per half-year — see Activity)

Vocabulary introduced today

- *ayana* (IAST: *ayana*; lit. “course, path”) — a half-year course of the Sun, either northward or southward.
- *uttarāyaṇa* (IAST: *uttara-āyaṇa*; lit. “northern course”) — the half-year from the winter solstice (~21 December) to the summer solstice (~21 June). The Sun’s noon altitude *rises* in the Northern Hemisphere over this period.
- *dakṣiṇāyana* (IAST: *dakṣiṇa-āyaṇa*; lit. “southern course”) — the other half-year, ~21 June to ~21 December. The Sun’s noon altitude *falls* in the Northern Hemisphere.
- **Solstice** — Latin *sol-stat*, “Sun stands still.” On the solstice day, the Sun’s noon altitude reaches its highest or lowest value of the year, then begins reversing direction. Sanskrit name: *ayana-saṁkrānti*.

Lesson flow

Warm-up (5 min)

- Open the calendar. The four marks from yesterday — March 21, June 21, September 23, December 21 — are visible.
- Prompt: *“Today (insert today’s date) — which ‘turning point’ is closest? And which side of it are we on?”*
- Take 2 responses. They’ll likely give the right answer with help.

Core (20 min)

1. **Two halves of the year.** Sketch on the board:

```
                 21 June  (summer solstice)
                     ↑
                     |
                     |  Uttarāyaṇa (northward)
                     |  — Sun's noon altitude rising
                     |
 21 Dec ←------------+-------------→ (other direction)
 (winter                            |
```

```
      solstice)                                  |  Dakṣiṇāyana (southward)
                                                 |  — Sun's noon altitude
falling
                                                 |
                                                 ↓
                                              21 June
```

No, draw it as a clean cycle:

```
    Dec 21 ── Jan ── Feb ── Mar ── Apr ── May ── Jun 21
      |                                             |
      |   UTTARĀYAṆA (northward / noon rising)      |
      |                                             |
    Jun 21 ── Jul ── Aug ── Sep ── Oct ── Nov ── Dec 21
      |                                             |
      |   DAKṢIṆĀYANA (southward / noon falling)    |
      |                                             |
```

2. **What does "northward" mean exactly?**
   - The Sun's **noon position** in the southern sky shifts NORTH a little each day during *uttarāyaṇa*.
   - In northern India (e.g. Delhi, latitude ~28°N), the noon Sun is **very low in the south** in late December and **almost overhead** in late June.
   - "Almost overhead" means the noon shadow of a vertical stick is at its shortest.
3. **Paraphrase from the Sūrya-siddhānta** (Middle-band citation requirement):

   The Indian astronomical text **Sūrya-siddhānta** (compiled in its current form between the 5th and 10th centuries CE) describes the year as having two *ayanas* — *uttarāyaṇa* and *dakṣiṇāyana* — and gives mathematical formulas for the Sun's daily motion in each. — paraphrased from Sūrya-siddhānta, chapter on solar motion (see `sources.md`)
4. **Tropic of Cancer and Tropic of Capricorn** (linking to Geography):
   - The latitude at which the Sun is *directly overhead at noon on the June solstice* = 23.5° N = **Tropic of Cancer**. (Cancer = the Latin/Greek name for *Karka rāśi*. The Sun used to be in Karka at the June solstice — Day 7 will explain why "used to be.")
   - The latitude at which the Sun is directly overhead on the December solstice = 23.5° S = **Tropic of Capricorn**. (Capricorn = *Makara*. Same caveat.)
   - India straddles the Tropic of Cancer — it runs roughly through Bhopal, Ranchi, and Kolkata.

Activity (15 min) — Calendar marking and formative quiz Part A

Part 1 (8 min) — calendar marking:

- On their marked calendar, students label each half:
  - From Dec 21 → Jun 21: write "**UTTARĀYAṆA**" in colour

    - From Jun 21 → Dec 21: write "**DAKṢIṆĀYANA**" in colour
- They also write the Sanskrit names next to the four turning-points:
    - Dec 21: *Uttarāyaṇa begins (winter solstice)*
    - Jun 21: *Dakṣiṇāyana begins (summer solstice)*
    - Mar 21 / Sep 23: *Viṣuva* — we'll meet these tomorrow.

Part 2 (7 min) — Formative Quiz Part A: see `quizzes/formative.md`. Collect.

Wrap-up (5 min)

- Exit prompt: *"Today (today's date) — are we in uttarāyaṇa or dakṣiṇāyana? Justify in one sentence."*
- Preview Day 4: *"Tomorrow — the equinoxes. The days when day and night are equal."*

Student-facing content

The Sun's year has two halves:

Half	Sanskrit name	When	What the Sun does (Northern Hemisphere)
Northward	*uttarāyaṇa*	~21 Dec → ~21 Jun	Noon altitude rises; days lengthen
Southward	*dakṣiṇāyana*	~21 Jun → ~21 Dec	Noon altitude falls; days shorten

The turning days are the **solstices** (in Sanskrit, *ayana-saṁkrāntis*).

The latitudes where the Sun is directly overhead at the two solstices have names — **Tropic of Cancer** (23.5° N, on the June solstice) and **Tropic of Capricorn** (23.5° S, on the December solstice). India straddles the Tropic of Cancer.

Homework

- Pick a vertical object near your home that you can see at noon (a flagpole, a tree, a building). For two days this week, at exactly 12:00 noon, look at its shadow. Is the shadow getting *shorter* or *longer*? In a sentence: which *ayana* are we in, and does your shadow observation match?

Differentiation

- **One tier up:** Indian latitude: ~8° N (Kanyākumāri) to ~35° N (Srinagar). For a place at 30° N, how high (in degrees from the horizon) is the noon Sun on the June solstice? On the December solstice? Hint: 90° − latitude ± 23.5°.
- **One tier down:** Focus on the two names. *Uttara* = north. *Dakṣiṇa* = south. *Ayana* = course. That's enough for today.

Teacher notes

- Don’t conflate *uttarāyaṇa* with the festival of *Makara Saṁkrānti* yet. Today’s *uttarāyaṇa* starts on **December 21** (the modern *sāyana* solstice). *Makara Saṁkrānti* falls on **~January 14** (the *nirayana* sidereal reckoning). They are off by ~24 days. Why? Day 7 unpacks it. Just hold the puzzle for now if a student raises it.
- The Sūrya-siddhānta paraphrase satisfies the Middle-band citation requirement for the module.

Citation(s) used in this lesson

- **Sūrya-siddhānta** (Sanskrit astronomical compendium, current critical text 5th–10th c. CE) — paraphrase of the two-*ayana* division of the solar year. See `sources.md` for the full reference.

Day 4 — The Equinoxes (Viṣuva)

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will define an equinox as the day when day and night are equal, identify the two equinoxes in the year, and explain — in terms of axial tilt — why the days are equal on those two dates.

Materials

- The Day 2 torch-globe setup
- A printable day-length graph (x-axis = months, y-axis = day length in hours; blank for the students to fill)
- Student notebooks

Vocabulary introduced today

- *viṣuva* (IAST: *viṣuva*; lit. “equal, balanced”) — an **equinox**, one of the two days a year when day and night have equal length everywhere on Earth (~12 hours each). Two of them: the **vernal/March equinox** (~21 March) and the **autumnal/September equinox** (~23 September).
- *viṣuvad-dina* — another classical name, lit. “the equal day.”

Lesson flow

Warm-up (5 min)

- Prompt: *“What does ‘equinox’ literally mean?”* — break it down: equi (equal) + nox (night). Latin for “equal night.” The Sanskrit *viṣuva* means the same.
- Recall yesterday’s diagram. Today’s focus is on the two SIDEWAYS positions of Earth (March and September).

Core (20 min)

1. **The geometry of an equinox.** Set up the torch-globe again:

    - Tilt the globe so the N pole leans neither toward nor away from the torch — it's "sideways."
    - Notice the **terminator** (the line between light and dark on the globe) — it now passes through both poles, dividing Earth into two equal hemispheres of day and night.
    - **Consequence:** every place on Earth gets 12 hours of day and 12 hours of night.

2. **Why "equal" everywhere?** The day length at any latitude depends on what fraction of that latitude-line is in daylight. On the equinox, the day/night terminator passes through both poles — so every latitude-line is exactly half lit, half dark.

3. **Two equinoxes per year:**

Date	Name	What happens after
~21 March	**Vernal / spring equinox** (*vasanta-viṣuva*)	Days get longer in N hemisphere → spring → summer
~23 September	**Autumnal / autumn equinox** (*śarad-viṣuva*)	Days get shorter in N hemisphere → autumn → winter

4. **Two equinoxes + two solstices = four turning points of the year.** Draw on board:

```
           ~21 June (summer solstice)
                 ☀ Sun highest at noon, longest day
                    (turning point — uttarāyaṇa ends, dakṣiṇāyana begins)
                    /
     ~23 Sept (autumn equinox / śarad-viṣuva)
           day = night everywhere
                    /
           ~21 Dec (winter solstice)
                 ☀ Sun lowest at noon, shortest day
                    (turning point — dakṣiṇāyana ends, uttarāyaṇa begins)
                    /
     ~21 Mar (spring equinox / vasanta-viṣuva)
           day = night everywhere
```

5. **A note on the dates:** The dates shift by ±1 day depending on which year (leap years, the slight unevenness of Earth's orbit). Modern almanacs give exact dates and times.

Activity (15 min) — The day-length graph

In pairs, students fill a blank graph:

- x-axis: months January to December
- y-axis: day length in hours (8 to 14)

The teacher provides a small table of approximate day lengths in Delhi (latitude 28.6° N):

Month	Day length (Delhi)
Jan	~10 h 25 min
Feb	~11 h 05 min
Mar	~12 h 00 min
Apr	~12 h 50 min
May	~13 h 30 min
Jun	~13 h 50 min
Jul	~13 h 40 min
Aug	~13 h 00 min
Sep	~12 h 10 min
Oct	~11 h 25 min
Nov	~10 h 40 min
Dec	~10 h 15 min

Students plot the points, connect them, and mark:

- The **two highest points** (close together in June) — summer solstice.
- The **two lowest points** (close together in December) — winter solstice.
- The **two crossings of the 12-hour line** — the equinoxes.

Discussion (last 3 min):

- Why is the difference between longest and shortest day only ~3.5 hours in Delhi, but ~16 hours in London (latitude 51° N)? *(Answer: the higher the latitude, the more extreme the day-length swing.)*
- What would this graph look like at the equator? *(Answer: nearly flat at 12 h all year.)*
- At the North Pole? *(Answer: 24 h of daylight in summer, 24 h of darkness in winter.)*

Wrap-up (5 min)

- Exit ticket: *"Why are equinoxes called 'equal'? Answer in one sentence."*
- Preview Day 5: *"Tomorrow — the 12 rāśis. The Sun moves through one each month."*

Student-facing content

Equinox = day equals night.

Sanskrit	Date	Position of Earth	Day length
vasanta-viṣuva	~21 March	tilt sideways	12 h day, 12 h night
śarad-viṣuva	~23	tilt sideways the other way	12 h day, 12 h night

Sanskrit	Date	Position of Earth	Day length
	September		

Together with the two **solstices** (June and December), these are the four turning-points of the year.

The Sun crosses the celestial equator on each equinox — moving from the Southern celestial hemisphere into the Northern (in March), or vice versa (in September).

Homework

- Compare the sunrise / sunset times in your local newspaper or weather app TODAY with the times exactly six months from now (you can look up future-six-months online or in a printed almanac). Calculate the difference in day length. One paragraph.

Differentiation

- **One tier up:** What is the "true" length of the solar year? Why is it 365.25 days and not 365 exactly? How does the calendar handle this? (Answer involves leap years; the Gregorian system; the Indian *adhikamāsa* in lunisolar context.)
- **One tier down:** Just remember: two equinoxes, two solstices, four turning-points. Mark all four on a calendar.

Teacher notes

- Some students will object: "the newspaper says sunrise was at 6:23, sunset at 18:31 — that's 12 hours and 8 minutes, not 12 hours exactly." That's because the Sun is a **disc** (not a point) and "sunrise" is when the **top** of the disc clears the horizon, and "sunset" is when the **last** of the disc disappears. So there's a small offset (~5–10 min depending on latitude). Also, atmospheric refraction bends the Sun's light, making it visible slightly before geometric sunrise. Accept the observation and explain it briefly.
- The day-length graph activity often produces the "aha" moment — students *see* the year's shape.

Citation(s) used in this lesson

- Day-length figures are approximate values for Delhi (28.6° N), computable from standard spherical-astronomy formulas. Verifiable in any almanac.

Day 5 — The 12 Rāśis

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will recite the 12 *rāśi* names in order in Sanskrit, match each to its Greek/English equivalent, and identify the approximate calendar month during which the Sun is in each *rāśi* (using the *nirayana* / sidereal system).

Materials

- A large printed circle divided into 12 equal segments (the *rāśi*-wheel template)
- Coloured pencils / sketch pens
- A printed reference table of the 12 *rāśi* names with English translations (from the glossary)
- Student notebooks

Vocabulary introduced today

- *rāśi* (IAST: *rāśi*; lit. "heap, mass; a sign of the zodiac") — one of 12 thirty-degree-wide segments of the **ecliptic** (the Sun's annual path on the sky). The Sun is in each *rāśi* for about 30 days.

Lesson flow

Warm-up (5 min)

- Prompt: *"What's your 'star sign' according to newspaper horoscopes?"* — students call out.
- Teacher: *"Those English names — Aries, Taurus, Gemini — they have Sanskrit equivalents. The Indian tradition has 12 rāśis, and the Sun visits one each month. Today we learn them."*

Core (20 min)

1. **The ecliptic is divided into 12 equal segments.** Each segment is 360° / 12 = **30°** of the sky. The Sun crosses each in about 30 days (well, 30.44 on average — see Day 6 footnote).

2. **The 12 *rāśi* names in order.** Write on the board:

#	Sanskrit (IAST)	Greek/Latin/ English	What the symbol depicts
1	*Meṣa*	Aries	Ram
2	*Vṛṣabha*	Taurus	Bull
3	*Mithuna*	Gemini	Twins
4	*Karka / Karkaṭa*	Cancer	Crab
5	*Siṁha*	Leo	Lion
6	*Kanyā*	Virgo	Maiden
7	*Tulā*	Libra	Scales
8	*Vṛścika*	Scorpio	Scorpion
9	*Dhanus*	Sagittarius	Bow / Archer
10	*Makara*	Capricorn	Sea-goat
11	*Kumbha*	Aquarius	Water-pot
12	*Mīna*	Pisces	Fishes

Notice: the Sanskrit and Greek images are *nearly identical*. The bull, twins, crab, lion, scales, scorpion, archer, fishes — same animals. This is not a coincidence; the two systems were in contact during the Hellenistic period (last few centuries BCE) and there was real exchange. Day 9 will compare them in detail.

3. **When is the Sun in each rāśi?** Roughly (in the *nirayana* / sidereal system used by traditional Indian almanacs):

Rāśi	Sun is here approximately
Meṣa	14 Apr → 14 May
Vṛṣabha	14 May → 14 Jun
Mithuna	14 Jun → 16 Jul
Karka	16 Jul → 16 Aug
Siṁha	16 Aug → 16 Sep
Kanyā	16 Sep → 17 Oct
Tulā	17 Oct → 16 Nov
Vṛścika	16 Nov → 15 Dec
Dhanus	15 Dec → 14 Jan
Makara	14 Jan → 13 Feb
Kumbha	13 Feb → 14 Mar
Mīna	14 Mar → 14 Apr

Each crossing is a *saṁkrānti* (Day 6).

4. **Paraphrase from the Sūrya-siddhānta** (re-anchoring the citation):

> The Indian astronomical text **Sūrya-siddhānta** divides the ecliptic into 12 equal *rāśis* of 30° each and gives the Sun's daily angular speed — about 0°59′8″ per day on average — which corresponds to one full circuit in ~365.25 days. — paraphrased from Sūrya-siddhānta, chapter on the Sun's motion (see `sources.md`)

Activity (15 min) — Begin the *rāśi*-wheel + formative quiz Part B

Part 1 (10 min) — start the wheel:

- Distribute the *rāśi*-wheel template (a circle divided into 12 wedges, each labelled with a number).
- Students write the Sanskrit name in each wedge in clockwise order (starting from *Meṣa* at the top).
- Below the Sanskrit name, the English / Greek name.
- Below that, the calendar month-range (e.g. *Meṣa* — 14 Apr → 14 May).
- They will finish coloring and decorating during Days 8–9 project work.

Part 2 (5 min) — Formative Quiz Part B: see `quizzes/formative.md`. Collect.

Wrap-up (5 min)

- Choral recitation: students read the 12 names aloud together, twice.
- Preview Day 6: *“Tomorrow — saṁkrānti. What the Sun does on the day it crosses from one rāśi to the next. There are ~12 saṁkrāntis per year. The most famous one is* ***Makara Saṁkrānti*** *— around 14 January.”*

Student-facing content

The ecliptic is divided into 12 equal segments — the *rāśis*.

Each *rāśi* is 30° of sky. The Sun spends ~30 days in each, and ~365 days to complete the cycle.

The 12 names, in order, are:

Meṣa, Vṛṣabha, Mithuna, Karka, Siṁha, Kanyā, Tulā, Vṛścika, Dhanus, Makara, Kumbha, Mīna.

They map directly onto the Greek/English: Aries, Taurus, Gemini, Cancer, Leo, Virgo, Libra, Scorpio, Sagittarius, Capricorn, Aquarius, Pisces.

Homework

- Memorise the 12 *rāśi* names in order. Use a phone-recording trick: record yourself saying them aloud, listen back twice a day. Test yourself by saying them without looking, in the order Meṣa → Mīna.
- Bonus: which *rāśi* is the Sun in right now? (Check the *nirayana* table; today’s date determines the answer.)

Differentiation

- **One tier up:** The *nirayana* table here is based on the Lahiri *ayanāṁśa* (~24° offset from the equinox). If you use the *sāyana* (tropical) system instead — used by Western horoscope columns — the Sun is in *Meṣa* / Aries from **21 March** to **20 April**, not 14 April to 14 May. Why the difference? Day 7 explains. (Hint: precession.)
- **One tier down:** Memorise just 4 names: *Meṣa* (Apr), *Karka* (Jul), *Tulā* (Oct), *Makara* (Jan). These are the four “anchors” — they fall near the solstices and equinoxes. The rest can come over the week.

Teacher notes

- Some students will get confused that their newspaper “Sun sign” doesn’t match the *nirayana* dates here. Reassure: this is expected — Western horoscope columns use the *sāyana* system; this table uses *nirayana*. They differ by ~24 days. Day 7 explains.
- The Sanskrit-Greek correspondence is genuinely striking and a good talking point. The historical reason: contact through the Hellenistic world after Alexander’s campaigns, with bidirectional flow.

Citation(s) used in this lesson

- **Sūrya-siddhānta** — paraphrase of the 12-*rāśi* division and the Sun’s mean angular speed. See `sources.md`.

Day 6 — Saṁkrānti

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will define *saṁkrānti* as the Sun's transit from one *rāśi* to the next, name **Makara Saṁkrānti** and **Karka Saṁkrānti** as the two solstice-related *saṁkrāntis*, and locate all 12 *saṁkrāntis* on a calendar.

Materials

- Yesterday's *rāśi*-wheel (work-in-progress)
- The marked calendar from Days 2–3
- A list of regional *Makara Saṁkrānti* names (Pongal, Lohri, Bihu, Maghi, Uttarayan)
- Activity 02 worksheet

Vocabulary introduced today

- *saṁkrānti* (IAST: *saṁ-krānti*; lit. "transit, crossing") — the moment the Sun crosses from one *rāśi* into the next. There are ~12 *saṁkrāntis* per year, one for each *rāśi*-boundary.
- *Makara Saṁkrānti* (IAST: *makara saṁkrānti*) — the *saṁkrānti* around **14 January** when the Sun enters *Makara* (Capricorn). Traditionally marks the start of *uttarāyaṇa* in the *nirayana* reckoning.
- *Karka Saṁkrānti* (IAST: *karka saṁkrānti*) — the *saṁkrānti* around **16 July** when the Sun enters *Karka* (Cancer). Traditionally marks the start of *dakṣiṇāyana* in the *nirayana* reckoning.

Lesson flow

Warm-up (5 min)

- Prompt: *"Have you ever celebrated **Pongal**, **Lohri**, **Bihu**, or **Uttarayan**?"* Likely many hands. *"All of these fall around the same date — January 14. Why?"*
- Take 2 responses. Then: *"They're all names for the same astronomical event: **Makara Saṁkrānti**. Today we learn what that is, astronomically."*

Core (20 min)

1. **Definition.** A *saṁkrānti* (transit) is a **moment**, not a day — the exact instant when the Sun's ecliptic longitude crosses a multiple of 30°, leaving one *rāśi* and entering the next. The *day* on which this happens is named after the *rāśi* the Sun is entering.

2. **The 12 *saṁkrāntis* in order:**

#	*Saṁkrānti*	Sun enters	Approximate date
1	*Meṣa Saṁkrānti* (also *Mahā Viṣuva* / Tamil *Puthandu*)	Meṣa (Aries)	14 April

#	*Saṁkrānti*	Sun enters	Approximate date
2	*Vṛṣabha Saṁkrānti*	Vṛṣabha (Taurus)	14 May
3	*Mithuna Saṁkrānti*	Mithuna (Gemini)	14/15 June
4	**Karka Saṁkrānti**	Karka (Cancer)	16/17 July
5	*Siṁha Saṁkrānti*	Siṁha (Leo)	16/17 August
6	*Kanyā Saṁkrānti*	Kanyā (Virgo)	16/17 September
7	*Tulā Saṁkrānti*	Tulā (Libra)	17/18 October
8	*Vṛścika Saṁkrānti*	Vṛścika (Scorpio)	16/17 November
9	*Dhanus Saṁkrānti*	Dhanus (Sagittarius)	15/16 December
10	**Makara Saṁkrānti**	Makara (Capricorn)	14/15 January
11	*Kumbha Saṁkrānti*	Kumbha (Aquarius)	12/13 February
12	*Mīna Saṁkrānti*	Mīna (Pisces)	14/15 March

3. **Two *saṁkrāntis* are especially significant:**
   - **Makara Saṁkrānti** (~14 January) — entry into *Makara*. In the *nirayana* system, this is the traditional start of *uttarāyaṇa*. Across India:
     - **Pongal** in Tamil Nadu (4-day harvest festival)
     - **Lohri** in Punjab and Haryana (bonfire eve)
     - **Bihu** in Assam (Magh Bihu)
     - **Maghi** in Punjab (Sikh new year)
     - **Uttarayan** in Gujarat (kite festival)
     - **Khichdi** in Uttar Pradesh and Bihar
     - **Suggi** in Karnataka
   - **Karka Saṁkrānti** (~16 July) — entry into *Karka*. In the *nirayana* system, this is the traditional start of *dakṣiṇāyana*. Celebrated as **Dakṣiṇāyana day** in parts of Maharashtra and as the **Rath Yatra** preparations in Odisha (loosely tied).
4. **Reference to the Bṛhat Saṁhitā** (Middle-band paraphrase):

   Varāhamihira (6th c. CE), in his encyclopedic *Bṛhat Saṁhitā*, dedicates chapters to the *saṁkrāntis* — listing their dates, their seasonal markers, and (in his cultural context) their religious significance. The astronomical core — the Sun crossing a *rāśi*-boundary — is what we extract here. — paraphrased from *Bṛhat Saṁhitā*, chapter on solar transits (see `sources.md`).

Activity (15 min) — Activity 02 *Makara Saṁkrānti* explainer

See `activities/activity-02-makara-sankranti.md` for the full sheet. In summary:

- In small groups of 3–4, students prepare a 2-minute explanation of *Makara Saṁkrānti* answering:
  - What is the astronomical event?
  - What date does it fall on?
  - What regional festivals celebrate it?

- What does it traditionally mark?

- Each group presents to one neighbouring group (pair-share). Last 3 minutes: one group presents to the full class.

Wrap-up (5 min)

- Exit prompt (verbal): *"Today is (today's date). Which saṁkrānti is closest? Past or upcoming?"*
- Preview Day 7: *"Tomorrow — the BIG question.* ***Why is Makara Saṁkrānti on 14 January and not on 21 December (the actual winter solstice)?*** *It's a 24-day gap. There's a reason."*

Student-facing content

Saṁkrānti = the Sun crosses into a new *rāśi*. There are 12 per year.

The two most important:

- **Makara Saṁkrānti** (~14 January) — Sun enters *Makara* (Capricorn). Marks the start of *uttarāyaṇa* in the *nirayana* (sidereal) system. Celebrated as Pongal / Lohri / Bihu / Maghi / Uttarayan across India.
- **Karka Saṁkrānti** (~16 July) — Sun enters *Karka* (Cancer). Marks the start of *dakṣiṇāyana*.

Each *saṁkrānti* is a calendar moment, named after the *rāśi* the Sun is entering. Local festivals attach to many of them.

Homework

- Add the 12 *saṁkrānti* dates to your *rāśi*-wheel — write the approximate date at each *rāśi*-boundary.
- If your family celebrates one of the *Makara Saṁkrānti* festivals (Pongal, Lohri, Bihu, etc.), ask one elder how they celebrate, and write 3 sentences. If your family doesn't celebrate any of them, write 3 sentences from a classmate's account.

Differentiation

- **One tier up:** Each *saṁkrānti* has a religious / cultural significance attached in traditional almanacs. List the 12 and their associated traditional names from a *pañcāṅga* or Wikipedia. Note: this is cultural-historical knowledge, not astronomy — keep the two separate.
- **One tier down:** Memorise just two — *Makara* (January) and *Karka* (July). That's enough for Day 7.

Teacher notes

- Resist the urge to "explain Day 7" early. Hold the puzzle: *Makara Saṁkrānti* on Jan 14 vs winter solstice on Dec 21 = 24-day gap. Let students sit with the gap; tomorrow we resolve it.
- Be ready for a religious-festival drift. Acknowledge briefly that these are real festivals; redirect to the astronomy.

- The Varāhamihira reference is paraphrased — don't pretend to quote.

Citation(s) used in this lesson

- **Varāhamihira, *Bṛhat Saṁhitā*** (6th c. CE) — paraphrased reference to the chapters on *saṁkrāntis*. See `sources.md`.

Day 7 — Sāyana vs Nirayana

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will distinguish the **sāyana** (tropical) and **nirayana** (sidereal) zodiacs, explain that the gap between them — the **ayanāṁśa** — is ~24° today, and link this gap to **precession** (the slow wobble of Earth's axis).

Materials

- Whiteboard
- Two transparent circles printed on overhead-projector film or paper-cutouts: each shows the 12 *rāśis* arranged around a circle
- A long printed strip showing the year (Jan to Dec) with the two solstice markers
- Student notebooks

Vocabulary introduced today

- *sāyana* (IAST: *sāyana*; lit. "with-ayana / tropical") — the reference frame fixed to the equinoxes. The Sun enters *Meṣa* (Aries) on the **March equinox** by definition. The seasons stay locked to this system.
- *nirayana* (IAST: *nir-ayana*; lit. "without-ayana / sidereal") — the reference frame fixed to the **fixed stars**. The Sun enters *Meṣa* when it physically passes a fixed point against the background stars.
- *ayanāṁśa* (IAST: *ayana-aṁśa*; lit. "precessional portion") — the angular offset between the two systems, ~24° today.
- **Precession** — the slow wobble of Earth's rotation axis. It traces a cone in space over ~25,800 years. Because of this wobble, the position of the equinoxes drifts among the stars at ~50 seconds of arc per year (~1° per 72 years).

Lesson flow

Warm-up (5 min)

- Recall the puzzle: *Makara Saṁkrānti* is ~14 January but the winter solstice is ~21 December — a 24-day gap.
- Prompt: *"24 days. Why 24?"*
- Take 2 guesses. Park them. *"Today we resolve it."*

Core (25 min)

1. **Two ways to define the start of the zodiac.**

   Imagine the great circle of the ecliptic divided into 12 *rāśis*. **Where does *Meṣa* (the first *rāśi*) begin?**

   - **Option A (*sāyana* / tropical):** *Meṣa* begins at **the March equinox point**. This makes the Sun's position lock-step with the seasons. When the Sun is at 0° *Meṣa*, it's the March equinox — always. Modern Western astronomy uses this.
   - **Option B (*nirayana* / sidereal):** *Meṣa* begins at **a fixed point against the stars** (chosen near the star Spica / *Citrā* by Indian convention). When the Sun is at 0° *Meṣa*, it's physically passing that star-anchored boundary. Indian traditional *jyotiṣa* uses this.
2. **Why do they differ?** Because of **precession**.
   - Earth's rotation axis isn't perfectly fixed in space. It's a slow wobble — like a spinning top tilting in a circle.
   - One full wobble takes about **25,800 years**.
   - The equinox point (where the Sun crosses the equator going north) is defined by the axis — so as the axis wobbles, the equinox point **drifts** against the background stars at about **50 seconds of arc per year**, or **1° per 72 years**.
   - The drift goes *backwards* through the zodiac — from *Meṣa* into *Mīna*, then into *Kumbha*, and so on.
3. **The size of the gap right now.**
   - The two systems were aligned approximately **1700 years ago** (in the early centuries CE).
   - In that time, precession has drifted the equinox by **~24°** against the stars.
   - So the *sāyana* zodiac and the *nirayana* zodiac are now **24° apart**.
   - **24° / 360° × 365.25 days ≈ 24.3 days**. That's the gap.
   - **24-day gap = 24° offset = ~1700 years of precession.** That's why *Makara Saṁkrānti* is 24 days *after* the winter solstice.
4. **Demo with the two transparencies.**
   - Place the two *rāśi* circles on top of each other, both starting at the top.
   - Slide one circle 24° clockwise. Now both circles still have *Meṣa, Vṛṣabha, …* in order, but the boundaries are offset.
   - "If the Sun is at 0° *Meṣa* in the bottom (sāyana) circle on March 21, where is it in the top (nirayana) circle?" — answer: it's at 0° *Mīna* + 6° (since it's offset 24° backward). It enters *Meṣa* nirayana ~24 days later, on ~April 14.
5. **A summary table:**

Event	*Sāyana* date	*Nirayana* date	Gap
Sun enters *Makara* /	~21 December (= winter solstice)	~14 January (= Makara Saṁkrānti)	~24 days

Event	*Sāyana* date	*Nirayana* date	Gap
Capricorn			
Sun enters *Karka* / Cancer	~21 June (= summer solstice)	~16 July	~25 days
Sun enters *Meṣa* / Aries	~21 March (= March equinox)	~14 April	~24 days
Sun enters *Tulā* / Libra	~23 September (= September equinox)	~17 October	~24 days

6. **Which is "right"?** Neither — they're answering different questions.
   - *Sāyana* answers: "Where is the Sun relative to the seasons (equinoxes / solstices)?" — used by modern astronomy and the civil calendar.
   - *Nirayana* answers: "Where is the Sun against the actual background stars?" — used by traditional Indian almanacs.
   - **Both are internally consistent.** They diverge because the equinox drifts; whichever frame you fix, the other one moves.

Activity (10 min) — Compare-contrast worksheet

(See `activities/activity-03-sayana-nirayana.md` for the full sheet.)

In pairs, students fill a two-column table:

Question	*Sāyana* answer	*Nirayana* answer
Sun in *Meṣa* on …	~21 March	~14 April
Aligned with seasons?	Yes (by definition)	Drifts over centuries
Aligned with stars?	Drifts over centuries	Yes (by definition)
Used by modern astronomy?	Yes	No
Used by Indian *pañcāṅga*?	No	Yes
What changes over millennia?	Star background drifts	Season alignment drifts

Wrap-up (5 min)

- Exit ticket: *"Why is Makara Saṁkrānti on January 14 and not December 21? Answer in one sentence."*
- Acceptable answer: "Because the *nirayana* (sidereal) zodiac, used for *Makara Saṁkrānti*, is offset from the *sāyana* (tropical) zodiac, used for the solstice, by ~24° due to precession over ~1700 years."

- Preview Day 8: *"Tomorrow — build the project. Your rāśi-wheel will show BOTH systems."*

Student-facing content

Two zodiacs, one sky.

System	Anchored to	Used by
Sāyana (tropical)	The **equinoxes**	Modern astronomy, Western horoscopes
Nirayana (sidereal)	The **fixed stars**	Indian traditional almanacs

The two were aligned ~1700 years ago. Earth's slow axial wobble — **precession** — has drifted them apart by **~24°** since. That gap is called the **ayanāṁśa**.

24° ÷ 360° × 365.25 days ≈ **24 days**. That's why *Makara Saṁkrānti* (Jan 14, *nirayana*) is 24 days after the winter solstice (Dec 21, *sāyana*).

Homework

- Look up the *ayanāṁśa* value for today on any astronomy website or *pañcāṅga* app. Most sources will quote either the *Lahiri* value (the Government of India standard) or the *Krishnamurti* value. They differ by a few minutes of arc. Note both.
- Write 4 sentences: explain to a younger sibling why their newspaper horoscope sign (Western, *sāyana*) might be one *rāśi* "off" from a traditional *pañcāṅga* (Indian, *nirayana*).

Differentiation

- **One tier up:** When will the *sāyana* and *nirayana* zodiacs realign? Hint: when *ayanāṁśa* completes another ~336° (= 25,800 yrs × 336/360). Approximately when? (Answer: in ~24,000 years.) What was the *ayanāṁśa* in 1 CE? (Answer: very close to 0°.)
- **One tier down:** Just remember: ~24 days difference. *Sāyana* = with the seasons. *Nirayana* = with the stars. Don't worry about precession mechanics yet.

Teacher notes

- This is the densest day of the module. Plan to take questions — and to defer some to next class. The full theory of precession is the Day 8 *senior* topic; here we just name it.
- Some students will be upset that "the calendar is wrong." Reframe: neither calendar is wrong. Each is right *within its frame*. The drift is a real astronomical phenomenon.
- If you have only 35 minutes, drop the worksheet and use the time for questions and the demo.

Citation(s) used in this lesson

- Precession rate (~50″/year, ~25,800-year period) is a modern astronomy standard value (see `sources.md`). Lahiri *ayanāṁśa* is the Government of India standard adopted in the 1950s by the Calendar Reform Committee chaired by Meghnad Saha.

Day 8 — Project Session 1: Build the Rāśi-Wheel

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this session, every pair has a substantial draft of the *rāśi*-wheel project — labelled, coloured, with the four solstice/equinox markers and at least 6 of the 12 *saṁkrānti* dates filled in.

Materials

- The in-progress *rāśi*-wheel from Day 5 (one per pair)
- Coloured pencils, sketch pens, markers
- A4 / A3 chart paper for an optional larger wheel
- Glue, scissors, ruler
- The *sources.md* file printed for reference
- Project rubric (`assessments/rubric.md`) — printed and visible

Lesson flow

Warm-up (5 min)

- Recap the project brief (5 sentences): each pair builds a *rāśi*-wheel that shows BOTH the *sāyana* and *nirayana* dates for the four turning-points, the 12 *rāśi* names with English/Greek equivalents, and at least one paragraph linking it to axial tilt and seasons.
- Show the rubric (it should already be on the board). Quick Q&A: 2 minutes for any rubric clarifications.

Core (5 min)

The teacher explicitly walks through the "Meets" target for the project:

- 12 *rāśi* names in correct order ✓
- English / Greek equivalents matching ✓
- Approximate *nirayana* date ranges (e.g. Meṣa = 14 Apr – 14 May) ✓
- All four solstice / equinox markers labelled with both *sāyana* and *nirayana* dates ✓
- A 4-sentence written explanation of why we have seasons (axial tilt) ✓
- Citation to *Sūrya-siddhānta* OR *Bṛhat Saṁhitā* OR an NCERT chapter ✓

To **Exceed**, add ONE of: - A clearly-drawn precession diagram showing the wobble cone - A second wheel for Greek-origin star-name comparisons - A short paragraph on the cross-cultural Sanskrit-Greek-Babylonian zodiac correspondence

Build time (30 min)

Pairs work in silence first (10 min) to get the labels down, then collaborative (20 min) to add the explanatory text, colour, and diagrams.

Teacher circulates:

- For pairs stuck on labelling: hand them the printed reference table from Day 5.
- For pairs stuck on axial-tilt explanation: redirect to the Day 2 sketch in their notebook.
- For pairs racing ahead: ask “what would happen if the tilt were 0°?” or “how would a Babylonian astronomer label this wheel differently?”

Check-in (5 min)

Stop work at minute 40. Each pair holds up their current draft. The teacher does a quick visual scan and notes 2–3 pairs to circle back to first thing tomorrow.

- Exit prompt: *“What’s the ONE thing you still need to finish before tomorrow?”*
- Write it on a sticky note attached to the wheel.

Pair coaching prompts (for the teacher)

If you see …	Try saying …
One partner doing all the drawing	“Tell me what your partner is contributing. Where will their writing go?”
A wheel with English names only	“Where do the Sanskrit names go? Show me.”
Confusion about *sāyana* vs *nirayana*	“Pick the four turning-points. Write the December solstice date on one line. Now write the Makara Saṁkrānti date below it. They’re different — your job is to make that visible on the wheel.”
Beautiful art, no astronomy	“Where on the wheel does the reader learn that the seasons come from axial tilt? Add a sentence.”

Differentiation

- **One tier up:** Add a second concentric ring showing the Western horoscope (*sāyana*) date range for each *rāśi*. The reader should be able to compare the two rings at a glance.
- **One tier down:** Drop the *sāyana* / *nirayana* dual labelling. Just label one set of dates clearly. The *ayanāṁśa* concept appears only in the written paragraph.

Teacher notes

- Some pairs will spend the entire 45 minutes colouring and lose the astronomy. Intervene at minute 20: “show me your written paragraph on axial tilt.” If it doesn’t exist yet, stop the colouring.
- A surprising number of pairs will get the *rāśi* order slightly wrong (especially *Tulā* ↔ *Vṛścika*). Catch this early.

Homework

- Finish at home what didn’t get done in class. Bring back tomorrow ready to refine, not start.
- If the pair worked separately, agree (over WhatsApp or in person) who finishes what.

Day 9 — Project Session 2: Cross-Cultural Comparison and Refinement

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this session, every pair has a presentation-ready *rāśi*-wheel including a cross-cultural comparison element (Greek / Babylonian / Gregorian) and a rehearsed 3-minute presentation script.

Materials

- The in-progress *rāśi*-wheel from Day 8
- Reference printout: Sanskrit / Greek / Babylonian zodiac names side by side (provided below in *Cross-cultural reference*)
- Stopwatch / phone timer
- The rubric (`assessments/rubric.md`)

Cross-cultural reference

The 12 zodiac signs across three traditions:

#	Sanskrit (Indian, *rāśi*)	Greek / English	Babylonian (~700 BCE origins)
1	*Meṣa* (Ram)	Aries (Ram)	LU.ḪUN.GÁ (Hired Man)
2	*Vṛṣabha* (Bull)	Taurus (Bull)	GU.AN.NA (Bull of Heaven)
3	*Mithuna* (Twins)	Gemini (Twins)	MAŠ.TAB.BA (Great Twins)
4	*Karka* (Crab)	Cancer (Crab)	AL.LUL (Crab)
5	*Siṁha* (Lion)	Leo (Lion)	UR.GU.LA (Lion)
6	*Kanyā* (Maiden)	Virgo (Maiden)	AB.SIN (Furrow / barley ear)
7	*Tulā* (Scales)	Libra (Scales)	RIN (Balance)
8	*Vṛścika* (Scorpion)	Scorpio (Scorpion)	GIR.TAB (Scorpion)
9	*Dhanus* (Bow)	Sagittarius (Archer)	PA.BIL.SAG (Archer)
10	*Makara* (Sea-goat)	Capricorn (Sea-goat)	SUḪUR.MAŠ (Goat-fish)
11	*Kumbha* (Water-pot)	Aquarius (Water-bearer)	GU.LA (Great One / water-pourer)
12	*Mīna* (Fishes)	Pisces (Fishes)	DU.NU.NU (Tails)

The Indian, Greek, and Babylonian names are *strikingly similar* — the same animals (ram, bull, twins, crab, lion, scorpion, fishes) appear in all three. This is not accidental. The most likely historical explanation is **shared origin in Babylonian astronomy (~700 BCE)**, transmitted to Greece (post-Alexander, ~300 BCE) and to India (Hellenistic contact, last centuries BCE). The

12-fold ecliptic division and the imagery were absorbed into the Indian *jyotiṣa* tradition and merged with earlier Vedic star-lore (the 27 *nakṣatra* system, which is genuinely older and independently Indian).

The 27-*nakṣatra* system **is** distinctively Indian; the 12-*rāśi* system is the Babylonian-Greek-Indian common ground.

Lesson flow

Warm-up (5 min)

- Project the cross-cultural table on the board. Take 30 seconds to let students absorb.
- Prompt: *"What patterns do you see?"* — collect 2 observations.
- Likely observation: "they're all about the same animals." Confirm: yes — and this is a real historical clue about cultural exchange.

Core (5 min)

- One-paragraph summary on the board:
  - The 12-fold zodiac is a shared inheritance across Babylon, Greece, and India.
  - Indian *jyotiṣa* layered this on top of the older 27-*nakṣatra* system.
  - This means: when you study the *rāśis*, you are studying a 2700-year-old transcontinental mathematical-astronomical inheritance, not a purely Indian invention.
- This is **not** "Indians borrowed from the West" — Babylonian astronomy reached India before it reached Greece in some respects, and the Indian elaboration (the *nirayana* convention, the *ayanāṁśa* corrections) is a real intellectual contribution.

Refinement work (25 min)

Pairs refine their wheel:

- Add the cross-cultural element. Options:
  - A small inset showing the Babylonian name for ONE *rāśi* alongside the Sanskrit and Greek.
  - A single sentence on the wheel: "These 12 signs appear in similar form in Babylonian, Greek, and Indian astronomy — a 2700-year inheritance."
  - A short comparative note on Cancer / Karka with a footnote on the Babylonian name.
- Finalise the written explanation paragraph (4–6 sentences) covering: axial tilt, the two *ayanas*, the four turning-points, the *sāyana* / *nirayana* distinction, and one citation.
- Decide who-says-what in the 3-minute presentation.

Rehearsal (8 min)

In groups of two pairs (4 students), pairs rehearse their 3-minute presentation in front of each other.

The listening pair times the speakers and gives one piece of feedback:

- ONE thing that came across clearly
- ONE thing the audience didn't understand

Then they swap.

Wrap-up (2 min)

- Exit signal: each pair raises hand showing 1, 2, 3, 4, or 5 fingers — how ready they feel for tomorrow.
- The teacher notes any pair showing 1 or 2 and plans a 5-minute conference at the start of Day 10.

Differentiation

- **One tier up:** Each pair preparing a "stretch" closer: a 30-second factoid the audience will remember (e.g. "Precession means in ~13,000 years, the December solstice will be in *Karka*, not *Makara*"). Confirm the factoid is astronomically accurate before they include it.
- **One tier down:** Skip the Babylonian table; just mention "the zodiac is shared between India, Greece, and the older Babylonian tradition." That's enough for a "Meets" presentation.

Teacher notes

- A small but real risk: a pair invents a Sanskrit etymology to "explain" a Greek name. Catch this — don't let invented etymology into a presentation.
- The Babylonian names contain Akkadian / Sumerian cuneiform transliterations (LU.ḪUN.GÁ, etc.). Don't drill these — they're for visual comparison, not memorisation.
- Lateness on the wheel is acceptable as long as the presentation is rehearsed. Better an unfinished but well-explained wheel than a beautiful unstructured one.

Homework

- Final touches. By tomorrow morning, the wheel should be:
  - ☐ Labelled in Sanskrit and English / Greek
  - ☐ Coloured legibly
  - ☐ Showing all four turning-points with dual *sāyana* / *nirayana* dates
  - ☐ Including at least one cross-cultural reference
  - ☐ Accompanied by a 4–6 sentence explanation
  - ☐ Ready for a 3-minute presentation by both partners

Day 10 — Final Assessment + Showcase

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min (extendable to 60 if pair count > 8)

Learning objective

By the end of this session, the module is closed: the summative quiz is sat, the *rāśi*-wheel projects are presented to peers, and reflection slips are submitted.

Materials

- Printed summative quiz, one per student (`quizzes/summative.md`)
- The rubric (`assessments/rubric.md`)
- A wall / pinboard for the *rāśi*-wheel display
- A stopwatch / timer
- Reflection slips (paper)

Lesson flow

Summative quiz (20 min)

- Students sit individually for the printed quiz.
- 15 minutes of work + 5 minutes of buffer.
- Closed book.
- Submit on the way out for the showcase.

Showcase (20 min)

- Pairs present their *rāśi*-wheel.
- 3 minutes per pair (strict). 1 minute for Q&A.
- Teacher marks on the rubric during each presentation (5 min slot per pair).
- Audience: the rest of the class. Each non-presenting student writes ONE positive comment on a slip for the presenting pair (collected at the end).

If you have more than 8 pairs and 20 minutes is tight, split into two simultaneous showcase clusters (4 pairs each, in different corners), with the teacher rotating between them.

Reflection (5 min)

Each student fills the reflection slip:

1. What is ONE thing you'll remember about the Sun's movement six months from now?
2. What is ONE thing that didn't click for you in this module — that you want to come back to?
3. (Optional) Was there a "wait — that's surprising" moment? What was it?

Submit before leaving.

What gets graded today

Item	Marks	Where it lives
Summative quiz	25	`quizzes/summative.md`

Item	Marks	Where it lives
Project artefact	30	rubric `assessments/rubric.md`
Presentation (Day 10)	10	rubric `assessments/rubric.md`
Total	**65** (scaled to 40 for module total — see rubric)	

Reflection slips are not graded; they inform next module's design.

Teacher notes

- The 3-minute timer is strict. The discipline of compression is part of the learning.
- If a pair runs over, the teacher gently raises a hand at 3 minutes; if they keep going past 3:30, end the presentation politely.
- For pairs that struggled — don't grade harshly during the presentation. Mark on substance + clarity, not on stage presence.
- Pin all the wheels on the wall after the showcase. Leave them up for a week.

Wrap-up

- 30 seconds, teacher: *"Two weeks ago, none of you could name the 12 rāśis. Today, all of you can — and you understand why **Makara Saṁkrānti** is on January 14 and not December 21. That's astronomical literacy that took the Indian tradition centuries to build. You built it in 10 days. Well done."*
- Closing prompt: *"Tomorrow morning, when you walk to school, notice which way the Sun is rising. You'll find yourself looking for it. That's the habit this module was after."*

Citation(s) used in this lesson

- All citations were addressed in earlier days. No new citations introduced today.

Differentiation

- **One tier up:** Students who finish the quiz early can begin annotating their *rāśi*-wheel with one additional astronomical fact each — checked by the teacher before being added to the displayed wheel.
- **One tier down:** Students struggling with the quiz: allow a 5-minute extension if needed; do NOT delay the showcase.

Writing Prompts

Writing Prompts — Movement of the Sun (Middle)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 200-word response.

1. In 3–4 sentences, explain Movement of the Sun to a friend who hasn't taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Movement of the Sun with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Movement of the Sun? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Homework

Homework — Movement of the Sun (Middle)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — The Sun's Yearly Path** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Axial Tilt and the Seasons** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Uttarāyaṇa and Dakshiṇāyana** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — Equinoxes — Viṣuva** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Twelve Rāśis** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Saṁkrānti Moments** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Sāyana vs Nirayana** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Precession of the Equinoxes** · Write 2 sentences about today's concept. (~10 min)

9. **Day 9 — Cross-Cultural Solar Calendars** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Final Share and Assessment** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.